

University POLITEHNICA of Bucharest  
Faculty of Electronics, Telecommunications and Information Technology

# Room Type Classification in Indoor Images Using Vision Transformers

## Dissertation Thesis

Submitted in partial fulfillment of the requirements for the award of  
the degree of *Master of Science*  
in the field of *Electronics, Telecommunications and Information Technology*  
study program *Advanced Wireless Communications*

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## Academic Integrity Declaration

I hereby declare that the thesis entitled *Room Type Classification in Indoor Images Using Vision Transformers*, submitted to the Faculty of Electronics, Telecommunications and Information Technology, University POLITEHNICA of Bucharest, as partial fulfillment of the requirements for the degree of *Master of Science* in the field of *Electronics, Telecommunications and Information Technology*, study program *Advanced Wireless Communications*, is my own work and has not been previously submitted, in whole or in part, to any other institution for the award of a degree or qualification.

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Bucharest, June 2026.

Candidate: Lupu Silviu-George

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## List of Abbreviations

ViT = Vision Transformer  
CE = Cross-Entropy  
CL = Center Loss  
SSL = Semi-Supervised Learning  
CM = Confusion Matrix  
mAP = mean Average Precision

# Chapter 1

## Introduction

### 1.1 Motivation

### 1.2 Problem Statement

We address the problem of classifying an indoor image into one of six room categories: bathroom, kitchen, living room, bedroom, hall, and dining room.

### 1.3 Contributions

- Train and tune a transformer-based baseline for six-class room recognition.
- Integrate Center Loss to improve embedding compactness and class separability.
- Simulate a semi-supervised scenario using online pseudo-labeling of unlabeled images.
- Evaluate using overall accuracy, confusion matrices, and embedding-space visualizations.

## Chapter 2

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